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## Greater China Semiconductors | Asia Pacific

# AI Networking: Scale-up Technology Enables China's AI Supernodes

Following our [WAIC takeaways](#), we further examine China's emerging scale-up landscape, the role of optical interconnect, and implications for the domestic interconnect supply chain.

**AI Supernodes enabled by scale-up are becoming the focus of China's AI computing landscape.** Alongside our China [WAIC takeaways](#), we provide an in-depth follow-up study of China's scale-up technology development. As we wrote in [China's AI Accelerators – Who's Poised to Win?](#), Chinese AI GPUs are strong in optical networking and server rack design at the system level, although they remain constrained by wafer process technology at the chip level ([Exhibit 6](#)). These chip-level constraints are shifting competition in China's AI computing market from standalone chip specs to system solutions. WAIC 2026 featured fewer chip launches and more supernode solutions. Leading domestic accelerator vendors displayed rack-scale or multi-rack supernodes with scale-up domains of 64 accelerators or more, mostly enabled by proprietary scale-up technologies.

**China's scale-up landscape is rapidly broadening, led by proprietary fabrics but spanning multiple system architectures ([Exhibit 4](#)).** A year ago, our US semis team published [Semiconductors: Scaling the AI Opportunity: A Scale-Up Network Primer \(29 Aug 2025\)](#); China is now catching up quickly in scale-up technology. Huawei remains the domestic scale leader, with Atlas 950 using UnifiedBus 2.0 to connect 1,024 NPUs in the demonstrated configuration and designed to scale to 8,192 NPUs. Biren's next-gen BR2xx NPO architecture targets up to 1,024 GPUs through BLink 2.0. Sugon's scaleX640 connects 640 accelerators in a single rack. Moore Threads(not covered) MTT C256 supports 128 GPUs in one rack and 256 across two racks through MT-Link 2.0, while MetaX, Alibaba and Enflame support 64- to 128-accelerator domains through MetaXLink-E, ICN, and GCU-LARE, respectively.

**Stock implications:** We believe **Montage** (OW, covered by Daniel Yen) is set to benefit from higher PCIe interconnect content in larger, distributed supernodes. As systems scale across racks, denser and longer PCIe links among CPUs, xPUs, switches, NICs and peripherals should support demand for PCIe retimers and PCIe switch chips. We also expect China AI GPU vendors, such as **Hygon** (OW, covered by Daisy Dai) (along with its ecosystem partner Sugon), **Cambricon** and **Iluvatar** (both OW, covered by Charlie Chan), to introduce or leverage scale-up technology to improve their AI server rack performance.

(Continued below.)

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# China AI Interconnect - Scale-up

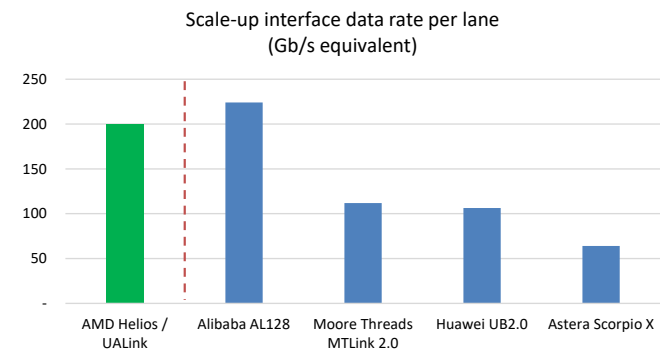
## Global AI server scale-up technology trend

### What is scale-up networking?

Scale-up networking refers to high-speed communication between GPUs/accelerators within the same server or rack. Interconnecting accelerators allow them to function as one large supercomputer, accessing the same memory and processing the same workload. This differs from scale-out networking, which connects multiple systems across a broader data center infrastructure.

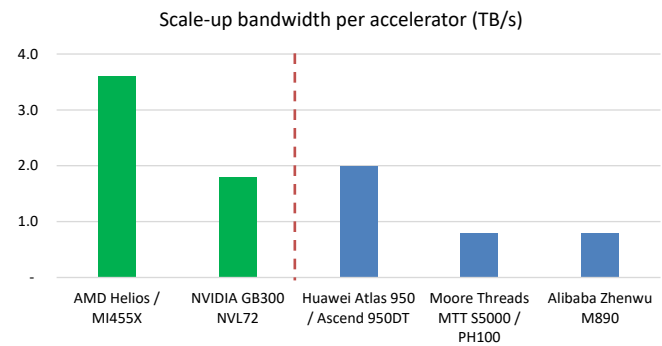
Montage's PCIe 6.x/CXL3.x x16 AEC has completed interoperability testing, and targets supernode and inter-rack deployments. Chinese systems deploying more xPUs per unit of compute could further increase PCIe content. Beyond standardized PCIe connectivity, multi-rack scale-up should also accelerate demand for domestic optical interconnect.

Exhibit 1: Scale-up interface data rate per lane



Source: Company data, Morgan Stanley Research

Exhibit 2: Scale-up bandwidth per accelerator



Source: Company data, Morgan Stanley Research

Exhibit 3: US AI networking scale-up technologies

| Technology  | NVLink5              | UALink 1.0    | SUE           | PCIe Gen 6                 | ICI                  | NeruoLink             |
|-------------|----------------------|---------------|---------------|----------------------------|----------------------|-----------------------|
| Purpose     | GPU-GPU; CPU-GPU     | xPU-xPU       | xPU-xPU       | Primarily CPU-XPU; xPU-xPU | TPU interconnect     | Trainium interconnect |
| Bandwidth   | 100 GB/s             | 100 GB/s      | 100 GB/s      | 64 GT/s                    | 50-100 c B/S         | not standardized      |
| Latency     | 10-20 nm             | <100 nm       | -             | -                          | varies               | varies                |
| Ecosystem   | NVIDIA (proprietary) | Open standard | Open standard | Open standard              | Google (proprietary) | Amazon (proprietary)  |
| Scalability | 576 GPUs             | 1,024 xPUs    | 1,024 xPUs    | varies                     | varies               | 64 Trainium           |

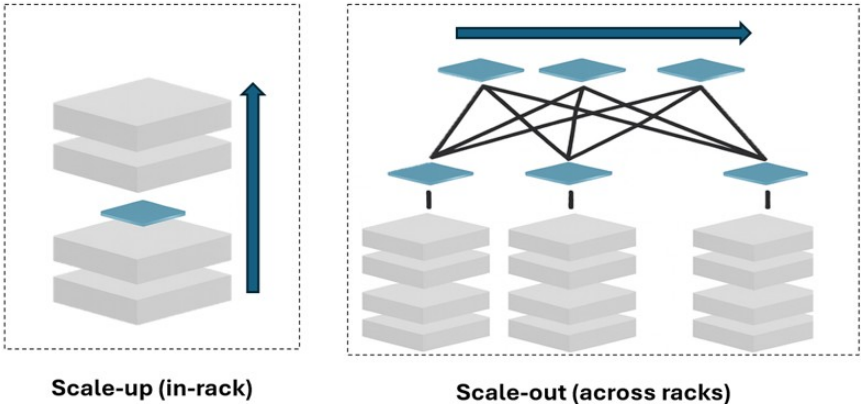
Source: Company data, Morgan Stanley Research

Exhibit 4: AI supernode scale-up solutions

| Scale-up tech summary                | NVIDIA                                             | AMD                                               | Asteralabs                              | Huawei                                                                                           | Moore Threads                                   | Sugon                                 | MetaX                                                   | Alibaba                                                  | Enflame                               | Enflame               | Biren                                                     | Biren                          |
|--------------------------------------|----------------------------------------------------|---------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------|---------------------------------------------------------|----------------------------------------------------------|---------------------------------------|-----------------------|-----------------------------------------------------------|--------------------------------|
| Supernode / accelerator name         | GB300 NVL72 (144 dies)                             | AMD Helios / MI455X                               | N/A                                     | Atlas 950 SuperPoD                                                                               | MTT C256 / MTT S5000, PH100                     | scaleX640                             | Xijing S600 / C600                                      | Panjiu AL128 / Zhenwu M890                               | Yunsui ES164-O                        | NPO optical prototype | LightSphere 128 / Bili 166L                               | BR2x NPO supernode (announced) |
| Scale-up solution                    | NVLink 5 + NVSwitch                                | UALink 1.0 / UALoE                                | Scorpio X-Series                        | UnifiedBus 2.0                                                                                   | MTLink 2.0                                      | scaleX fabric                         | MetaLink and MetaXLink-E                                | ALink system; M890 implements ICN + ICN Switch 1.0       | GCU-LARE                              | Undisclosed           | Undisclosed                                               | Blink 2.0                      |
| Technology foundation                | Proprietary                                        | Ethernet PHY-based, memory-semantic UALink fabric | PCIe Gen6                               | Proprietary                                                                                      | Proprietary                                     | Open multi-vendor system architecture | Proprietary                                             | Supports proprietary protocol and UALink                 | Proprietary                           | Undisclosed           | SiPh OCS                                                  | Proprietary                    |
| Singal lane speed                    | Physical-lane rate N/A; 100GB/s per NVLink 5 link  | 200GB/s per lane; 212.5GT/s raw signaling rate    | Undisclosed                             | 106.25Gb/s PAM4 under UBR2.0 specification                                                       | 112Gb/s per lane                                | Undisclosed                           | Undisclosed                                             | 112G/234G SerDes supported at AL128 platform level       | Undisclosed                           | Undisclosed           | Undisclosed                                               | Undisclosed                    |
| Total bandwidth per accelerator      | 1.8TB/s                                            | 3.6TB/s                                           | Undisclosed                             | 2TB/s                                                                                            | 800GB/s                                         | Undisclosed                           | Undisclosed                                             | 800GB/s                                                  | Undisclosed                           | Undisclosed           | Undisclosed                                               | Undisclosed                    |
| Total accelerator units in supernode | 72                                                 | 72                                                | Up to 80 per switch (merchant solution) | 1,024 NPUs demonstrated / 8,192 NPUs announced                                                   | 128 GPUs in one rack; 256 GPUs across two racks | 640                                   | 64 GPUs in 5600 rack; 128 GPUs through C600 MetaXLink-E | 128 per rack; 64 per disclosed full-bandwidth ICN domain | 64                                    | 512-plus              | 128                                                       | 1,024                          |
| Physical interface                   | Passive copper: NVLink cable-cartridge / backplane | Electrical/copper                                 | Electrical/copper                       | Hybrid: orthogonal cableless electrical interconnect intra-rack; optical interconnect inter-rack | All-copper Cable Tray                           | Electrical/copper                     | Cableless OEX electrical interconnect                   | Backplane-free orthogonal electrical; NPC interconnect   | Cableless OEX electrical interconnect | NPO optics            | SiPh optical circuit switching supplied by Lightelligence | NPO optics                     |
| Copper cable                         | Yes                                                | Yes                                               | Yes                                     | No (cableless)                                                                                   | Yes                                             | Undisclosed                           | No (cableless)                                          | Yes                                                      | No (cableless)                        | No                    | No                                                        | No                             |
| CPO                                  | No for scale-up (Yes for scale-out)                | No                                                | No                                      | Undisclosed                                                                                      | No                                              | Undisclosed                           | No                                                      | No                                                       | No                                    | No                    | No                                                        | No                             |
| NPO                                  | No                                                 | No                                                | No                                      | Undisclosed                                                                                      | No                                              | Undisclosed                           | No                                                      | No                                                       | No                                    | Yes                   | No                                                        | Yes                            |

Source: Company data, Morgan Stanley Research

Exhibit 5: Scale-up vs. scale-out networking



Source: Broadcom 2025 OCP Summit Keynote, Morgan Stanley Research

### China AI server scale-up development

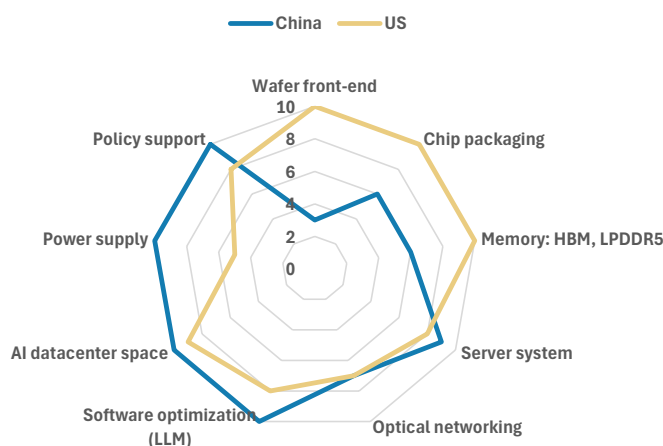
**In China, electrical interconnect remains the preferred option within racks, while optical becomes more important as scale-up domains extend across racks.** Short electrical links offer lower cost and simpler integration within boards, trays and single-rack systems. Moore Threads' C256 uses an all-copper cable tray design, while the orthogonal zero-cable architectures by Enflame and MetaX also remain electrical. Optical interconnection becomes more attractive across racks, where copper reach, signal loss and power consumption become constraints. Huawei's Atlas 950 SuperPoD extends UnifiedBus across racks within the same SuperPoD using optical interconnect.

**The transition toward optical connectivity may create demand for domestic switching and optical engine solutions designed for AI systems.** At WAIC 2026, Lightelligence (not covered) showcased its new NPO/CPO switch solutions (Exhibit 8, Exhibit 9), which pair the switch ASIC from a leading domestic vendor, which we believe is Centec Communications (not covered), with Lightelligence's optical engine, including the SiPh PIC and selected EICs such as the TIA. The NPO/CPO solutions shorten the high-speed

electrical path between the switch ASIC and optical engine, improving bandwidth density and power efficiency, while NPO retains greater serviceability than CPO. The same architecture can serve both scale-up and scale-out networks, with scale-up becoming increasingly relevant as tightly coupled accelerator domains extend across racks.

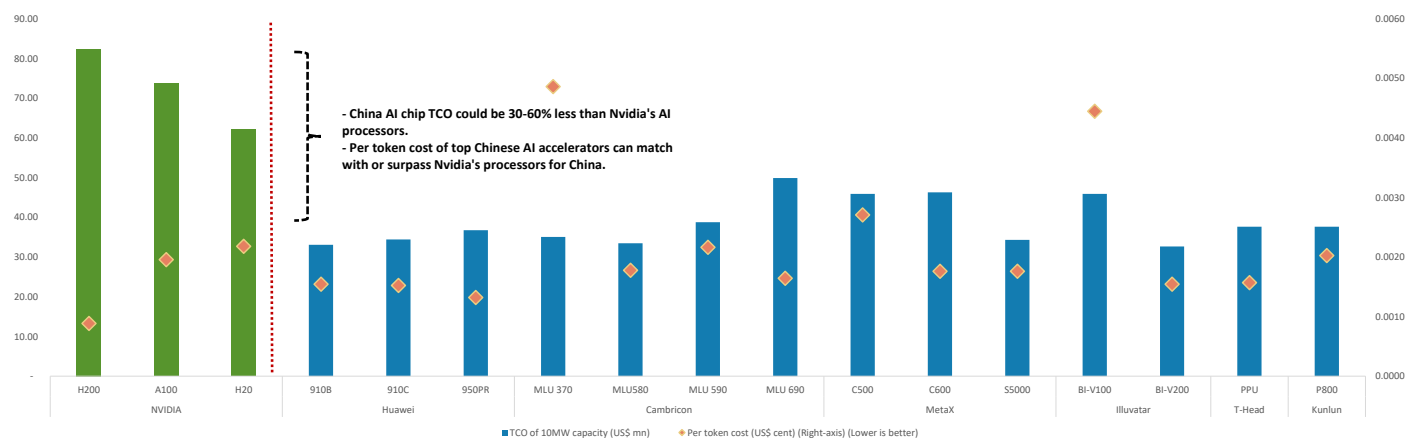
**More broadly, scale-up is emerging as the principal area of architectural differentiation, while scale-out remains the backbone of large-scale clusters.** System solutions generally use a specialized fabric to connect accelerators within a supernode, and Ethernet-based fabrics such as RoCE or UBoE, or Infiniband to connect multiple supernodes across a cluster. NVIDIA combines NVLink with InfiniBand or Spectrum-X Ethernet; AMD Helios pairs UALink with Ethernet scale-out; Huawei uses UnifiedBus within the SuperPoD, and UBoE or RoCE for scale-out between SuperPoDs.

**Exhibit 6:** Relative strengths of AI industries in the US and China - optical networking almost on par



Source: Morgan Stanley Research

**Exhibit 7:** Domestic chips have lower TCO and comparable per token cost (AI LLM inference) vs. NVIDIA's processors for China

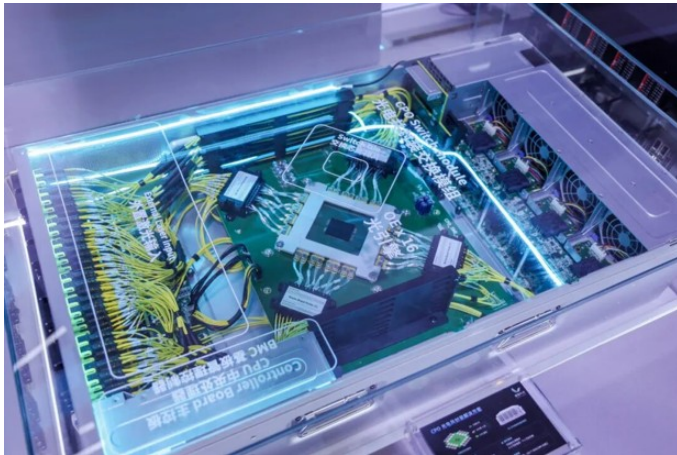


Source: Company data, Morgan Stanley Research estimates

## Optical interconnect is progressing towards scale-up system deployment

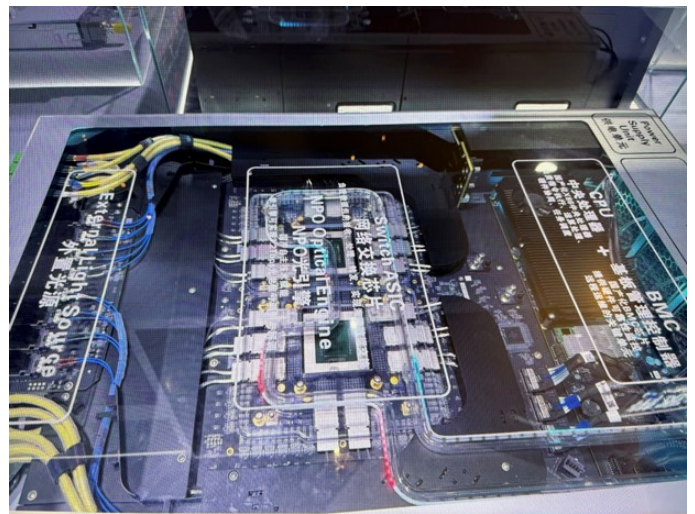
Enclave and Lightelligence previously demonstrated an xPU-CPO prototype that places the optical engine beside the accelerator, enabling direct optical output from the xPU and reducing the electrical distance between the compute silicon and optics. At WAIC 2026, Enclave displayed an NPO-equipped GPU-server solution targeting scale-up domains of 512 accelerators or more, developed in partnership with Lightelligence ([Exhibit 10](#)). Biren also unveiled a next-gen BR2xx-based NPO architecture designed to support a scale-up domain of up to 1,024 GPUs ([Exhibit 11](#)). Also, LightSphere X (jointly developed by Lightelligence, Biren, ZTE, and other partners) incorporates Lightelligence's SiPh interconnect and dOCS technology, enabling the accelerator domain to extend across racks and allowing the topology to be reconfigured according to workload requirements ([Exhibit 12](#)). These projects show domestic optical technology being integrated at both the accelerator optical-I/O layer and the distributed optical-switching layer, to support larger, multi-rack scale-up domains.

**Exhibit 8:** 51.2T CPO switch module (displayed by Lightelligence)



Source: Company data, Morgan Stanley Research

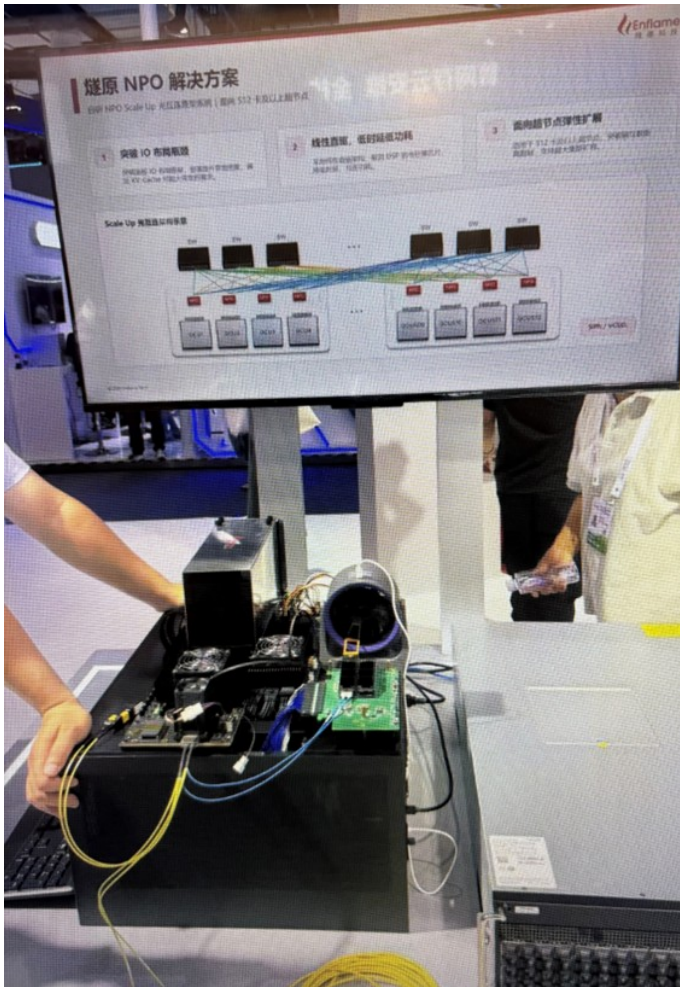
**Exhibit 9:** NPO switch module (displayed by Lightelligence)



Source: Company data, Morgan Stanley Research



**Exhibit 10:** Enflake's NPO optical interconnect solution for scale-up



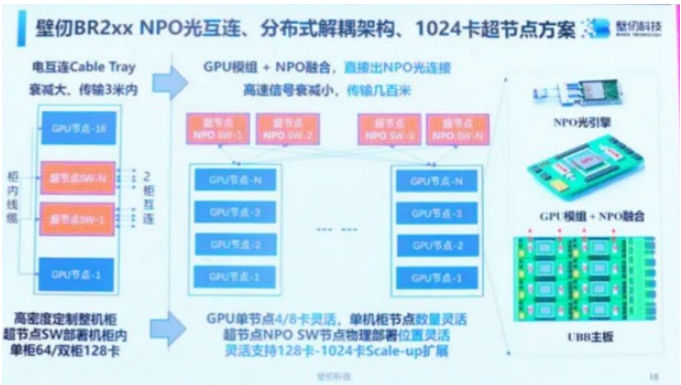
Source: Company data

**Exhibit 12:** LightSphere X (Lightelligence)



Source: Company data, Morgan Stanley Research

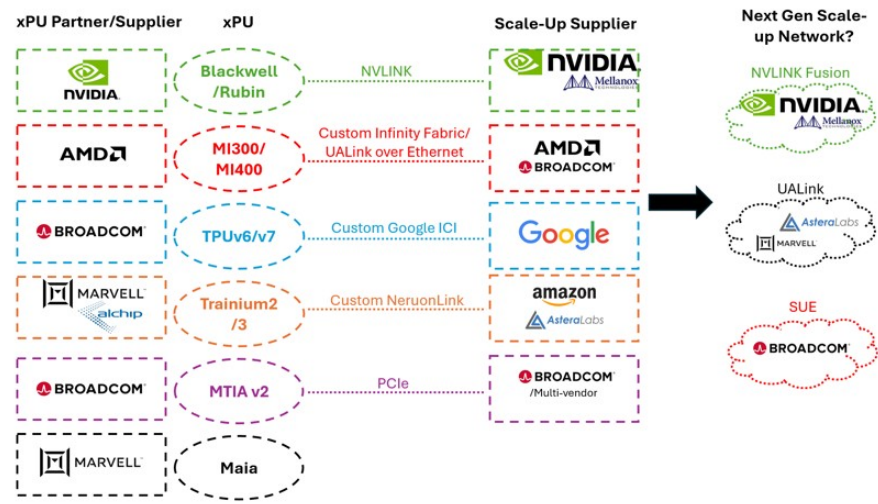
**Exhibit 11:** Biren's NPO optical interconnect solution for scale-up



Source: Company data

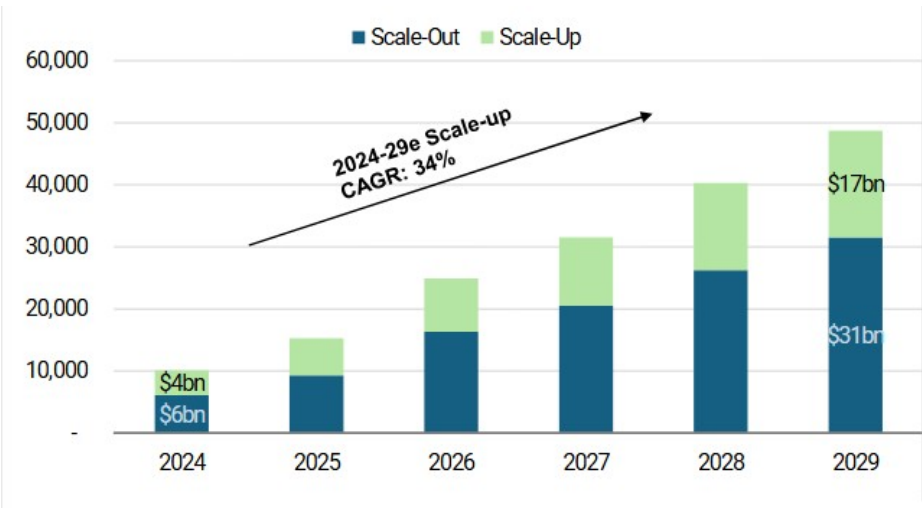
# US scale-up networking ecosystem and TAM forecast

Exhibit 13: Chinese CSPs' capex will be a key demand driver for Chinese AI GPUs



Source: Company data, Morgan Stanley Research

Exhibit 14: US semi : Scale-up TAM estimate (\$mn)



Source: Company data, Morgan Stanley Research (E) estimates

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### Montage Technology Co Ltd (6809.HK)

Base case, residual income model. Key assumptions:

- 8.4% CoE (1.2 beta, 3.0% risk-free rate, and 4.5% risk premium)
- 30% payout ratio
- 19.3% medium-term growth rate
- 4% terminal growth rate

These assumptions reflect cloud capex growth, DRAM interface technology migration, and Montage's strong position in China's datacenter semi localization. We then apply an exchange rate of 1.15 HKD:1 RMB, assuming no H-share discount vs the A-share.

#### Risks to Upside

- Faster-than-expected phase-out of US peers
- Faster-than-expected spec migration

#### Risks to Downside

- Weaker-than-expected cloud demand
- Slower-than-expected DRAM interface technology migration
- Delay in new product launches

### Montage Technology Co Ltd (688008.SS)

Base case, residual income model. Key assumptions:

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- 30% payout ratio
- 19.3% medium-term growth rate
- 4% terminal growth rate

We believe these assumptions are justified, given the cloud capex growth, DRAM interface technology migration, and Montage's strong position in China's datacenter semi localization.

#### Risks to Upside

- Faster-than-expected phase-out of US peers
- Faster-than-expected spec migration

#### Risks to Downside

- Weaker-than-expected cloud demand
- Slower-than-expected DRAM interface technology migration
- Delay in new product launches

### Hygon Information Technology Co., Ltd. (688041.SS)

We assume an 8.0% cost of equity (beta 1.05, risk-free rate 2.0% and risk premium 5.8%), a payout ratio of 50%, a medium-term growth rate of 25.0%, and a terminal growth rate of 5.0%, all of which are in line with other China AI chip companies under our coverage.

**Risks to Upside**

- Hygon further differentiates itself from local CPU and GPU competitors and gains significant market share in China's CPU and GPU markets
- China AI demand is stronger than expected
- Faster ramp-up of leading node capacity

**Risks to Downside**

- Intensified pricing competition among local GPU companies
- China AI demand is weaker than expected
- Slower yield improvement and capacity buildout at local leading node foundries

**Cambricon Technology Corporation (688256.SS)**

Key valuation assumptions underpinning our model include: an 8.4% cost of equity (derived from a beta of 1.06, risk-free rate of 2.0%, and equity risk premium of 6.0%), a long-term payout ratio of 40% (was 57%), a medium-term growth rate of 16%, and a terminal growth rate of 6.0%.

**Risks to Upside**

- Stronger-than-expected AI demand
- CSP order ramp-up
- Accelerating localization

**Risks to Downside**

- Capacity and yield constraints
- Customer concentration risk
- Slower technology iteration

**Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)**

Key valuation assumptions include:

- An 8.3% cost of equity (derived from a beta of 1.05, risk-free rate of 2.0%, and equity risk premium of 6.0%)
- A long-term payout ratio of 34% (was 33%)
- A medium-term growth rate of 16%
- A perpetual terminal growth rate of 6%

**Risks to Upside**

- Stronger-than-expected CSP orders
- Faster CUDA replacement with Iluvatar's software
- Expansion of overseas and domestic capacity

**Risks to Downside**

- Order ramp below expectations
- Escalation of sanctions
- Intensifying competition

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## Global Stock Ratings Distribution

(as of June 30, 2026)

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| Stock Rating<br>Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                         | Other Material Investment Services<br>Clients (MISC) |                          |
|--------------------------|-------------------|------------|----------------------------------|----------------|-------------------------|------------------------------------------------------|--------------------------|
|                          | Count             | % of Total | Count                            | % of Total IBC | % of Rating<br>Category | Count                                                | % of Total Other<br>MISC |
| Overweight/Buy           | 1544              | 42%        | 453                              | 49%            | 29%                     | 757                                                  | 44%                      |
| Equal-weight/Hold        | 1577              | 43%        | 390                              | 42%            | 25%                     | 769                                                  | 44%                      |
| Not-Rated/Hold           | 3                 | 0%         | 1                                | 0%             | 33%                     | 1                                                    | 0%                       |
| Underweight/Sell         | 544               | 15%        | 89                               | 10%            | 16%                     | 204                                                  | 12%                      |
| Total                    | 3,668             |            | 933                              |                |                         | 1731                                                 |                          |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

## Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

## Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

## Stock Price, Price Target and Rating History (See Rating Definitions)

Cambricon Technology Corporation (688256.SS) - As of 07/27/26 GMT in CNY  
Industry : Greater China Technology Semiconductors



Stock Rating History: 4/27/26 : O/A

Price Target History: 4/27/26 : 1065.77; 5/5/26 : 1342.28; 6/22/26 : 1528

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Hygon Information Technology Co., Ltd. (688041.SS) - As of 07/27/26 GMT in CNY  
Industry : Greater China Technology Semiconductors



Stock Rating History: 7/3/26 : O/A

Price Target History: 7/3/26 : 480

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

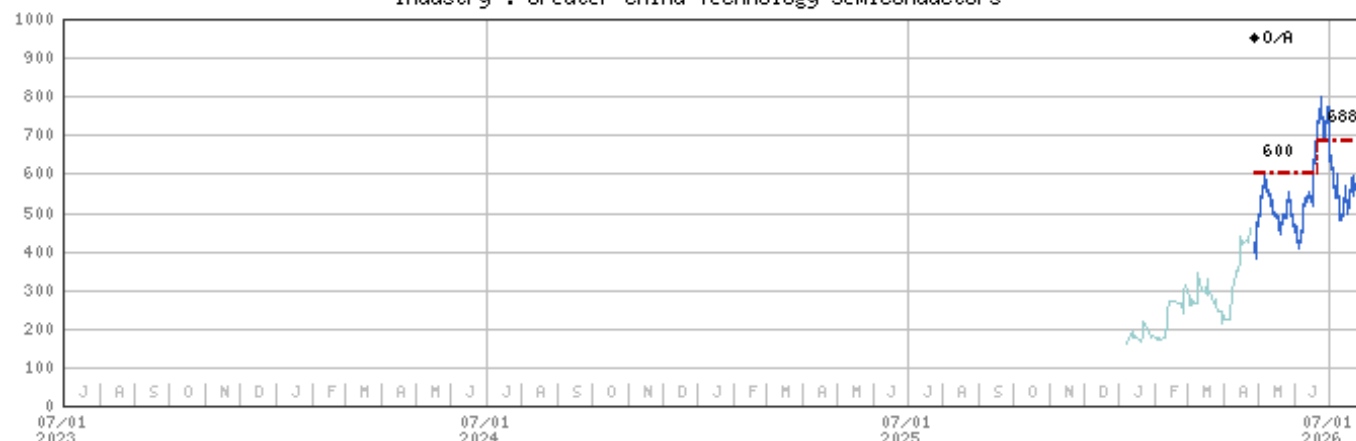
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.



Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK) - As of 07/27/26 GMT in HKD  
Industry : Greater China Technology Semiconductors



Stock Rating History: 4/27/26 : O/A

Price Target History: 4/27/26 : 600; 6/22/26 : 688

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target: -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Montage Technology Co Ltd (6809.HK) - As of 07/27/26 GMT in HKD  
Industry : Greater China Technology Semiconductors



Stock Rating History: 3/18/26 : O/A

Price Target History: 3/18/26 : 212; 4/28/26 : 310; 7/16/26 : 432

Source: Morgan Stanley Research Date Format: MM/DD/YY Price Target: -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

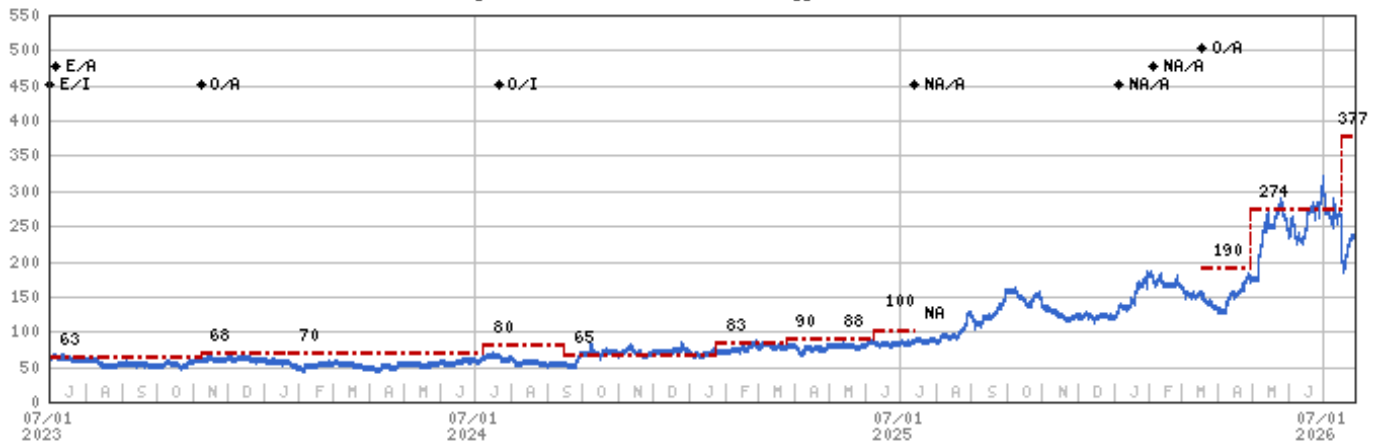
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Montage Technology Co Ltd (688008.SS) - As of 07/27/26 GMT in CNY  
Industry : Greater China Technology Semiconductors



Stock Rating History: 7/1/21 : E/I; 10/12/21 : E/C; 5/20/22 : U/C; 10/4/22 : U/A; 12/13/22 : O/A; 2/22/23 : O/I; 5/4/23 : E/I; 7/7/23 : E/A; 11/8/23 : O/A; 7/21/24 : O/I; 7/14/25 : NA/A; 1/5/26 : NA/A; 2/4/26 : NA/A; 3/18/26 : O/A

Price Target History: 5/5/21 : 55; 10/29/21 : 62; 11/16/21 : 80; 4/8/22 : 60; 5/20/22 : 52; 7/19/22 : 44; 10/12/22 : 38; 12/13/22 : 85; 1/11/23 : 78; 5/4/23 : 63; 11/8/23 : 68; 1/23/24 : 70; 7/8/24 : 80; 9/15/24 : 65; 1/24/25 : 83; 3/25/25 : 90; 5/6/25 : 88; 6/9/25 : 100; 7/14/25 : NA; 3/18/26 : 190; 4/28/26 : 274; 7/16/26 : 377

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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## INDUSTRY COVERAGE: Greater China Technology Semiconductors

| COMPANY (TICKER)                                     | RATING (AS OF) | PRICE* (07/27/2026) |
|------------------------------------------------------|----------------|---------------------|
| <b>Charlie Chan</b>                                  |                |                     |
| ACM Research Inc (ACMR.O)                            | O (03/07/2023) | US\$84.32           |
| Advanced Micro-Fabrication Equipment Inc (688012.SS) | O (11/06/2023) | Rmb399.77           |
| Advanced Wireless Semiconductor Co (8086.TWO)        | U (07/14/2025) | NT\$121.50          |
| Alchip Technologies Ltd (3661.TW)                    | O (05/14/2021) | NT\$3,460.00        |
| ASE Technology Holding Co. Ltd. (3711.TW)            | O (09/15/2024) | NT\$608.00          |
| Cambricon Technology Corporation (688256.SS)         | O (04/27/2026) | Rmb1,241.02         |
| Global Unichip Corp (3443.TW)                        | O (06/24/2026) | NT\$4,050.00        |
| GlobalWafers Co Ltd (6488.TWO)                       | E (05/19/2026) | NT\$1,065.00        |
| Gudeng Precision (3680.TWO)                          | O (11/25/2025) | NT\$450.50          |
| Hua Hong Semiconductor Ltd (1347.HK)                 | E (03/12/2026) | HK\$148.60          |
| Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)     | O (04/27/2026) | HK\$597.00          |
| King Yuan Electronics Co Ltd (2449.TW)               | O (03/03/2023) | NT\$276.00          |
| Maxscend Microelectronics Co Ltd (300782.SZ)         | U (01/11/2021) | Rmb72.90            |
| MediaTek (2454.TW)                                   | O (11/28/2025) | NT\$3,680.00        |
| MetaX Integrated Circuits (688802.SS)                | E (04/27/2026) | Rmb830.20           |
| Nanya Technology Corp. (2408.TW)                     | O (05/28/2026) | NT\$436.00          |
| NAURA Technology Group Co Ltd (002371.SZ)            | O (11/06/2023) | Rmb770.00           |
| OmniVision Integrated Circuits Group Inc (603501.SS) | E (11/17/2025) | Rmb95.31            |
| Phison Electronics Corp (8299.TWO)                   | E (02/25/2026) | NT\$1,820.00        |
| SG Micro Corp. (300661.SZ)                           | E (11/03/2025) | Rmb98.89            |
| Silergy Corp. (6415.TW)                              | U (05/19/2026) | NT\$449.00          |
| SMIC (0981.HK)                                       | O (10/21/2025) | HK\$70.65           |
| TSMC (2330.TW)                                       | O (02/07/2022) | NT\$2,350.00        |
| UMC (2303.TW)                                        | O (05/19/2026) | NT\$126.00          |
| Vanguard International Semiconductor (5347.TWO)      | E (01/14/2026) | NT\$157.00          |
| WIN Semiconductors Corp (3105.TWO)                   | U (07/14/2025) | NT\$355.00          |

**Daisy Dai, CFA**

|                                                      |                |            |
|------------------------------------------------------|----------------|------------|
| ASMP Ltd (0522.HK)                                   | O (07/24/2025) | HK\$161.60 |
| China Resources Microelectronics Limited (688396.SS) | U (03/02/2026) | Rmb60.74   |
| Empyrean Technology Co Ltd (301269.SZ)               | E (01/17/2025) | Rmb92.22   |
| Hangzhou Silan Microelectronics Co. Ltd. (600460.SS) | U (08/25/2025) | Rmb32.33   |
| Hygon Information Technology Co., Ltd. (688041.SS)   | O (07/03/2026) | Rmb313.59  |
| Innoscence (2577.HK)                                 | E (10/13/2025) | HK\$44.12  |
| JCET Group Co Ltd (600584.SS)                        | E (01/16/2026) | Rmb82.35   |
| Shanghai Fudan Microelectronics (1385.HK)            | O (03/07/2025) | HK\$24.88  |
| SICC Co Ltd (688234.SS)                              | O (03/20/2026) | Rmb100.00  |
| StarPower Semiconductor Ltd (603290.SS)              | E (05/14/2026) | Rmb84.88   |
| Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)  | U (01/10/2023) | Rmb66.00   |
| Universal Scientific Ind. (Shanghai) (601231.SS)     | O (11/05/2025) | Rmb24.77   |
| Yangjie Technology (300373.SZ)                       | O (06/10/2022) | Rmb93.22   |

**Daniel Yen, CFA**

|                                                    |                |               |
|----------------------------------------------------|----------------|---------------|
| AP Memory Technology Corp (6531.TW)                | O (07/11/2025) | NT\$773.00    |
| ASMedia Technology Inc (5269.TW)                   | U (10/03/2025) | NT\$1,385.00  |
| Aspeed Technology (5274.TWO)                       | O (06/09/2025) | NT\$14,695.00 |
| Egis Technology Inc (6462.TWO)                     | E (01/28/2026) | NT\$98.70     |
| Elan Microelectronics Corp (2458.TW)               | O (10/03/2025) | NT\$166.00    |
| Espressif Systems (688018.SS)                      | O (05/15/2023) | Rmb104.23     |
| GigaDevice Semiconductor Beijing Inc (603986.SS)   | O (05/15/2025) | Rmb434.03     |
| Macronix International Co Ltd (2337.TW)            | O (09/18/2025) | NT\$125.50    |
| Montage Technology Co Ltd (6809.HK)                | O (03/18/2026) | HK\$322.00    |
| Montage Technology Co Ltd (688008.SS)              | O (03/18/2026) | Rmb227.45     |
| Novatek (3034.TW)                                  | U (02/04/2026) | NT\$518.00    |
| Nuvoton Technology Corporation (4919.TW)           | U (11/10/2025) | NT\$132.50    |
| Parade Technologies Ltd (4966.TWO)                 | O (05/27/2026) | NT\$615.00    |
| Powerchip Semiconductor Manufacturing Co (6770.TW) | O (10/27/2025) | NT\$61.90     |
| Realtek Semiconductor (2379.TW)                    | E (01/30/2026) | NT\$762.00    |
| Shenzhen Goodix Technology Co Ltd (603160.SS)      | U (07/14/2025) | Rmb50.31      |
| Winbond Electronics Corp (2344.TW)                 | O (05/28/2026) | NT\$160.00    |
| WPG Holdings (3702.TW)                             | O (03/16/2026) | NT\$116.00    |
| WT Microelectronics Co. Ltd. (3036.TW)             | O (01/27/2026) | NT\$203.50    |

**Duan Liu**

|                                                 |                |           |
|-------------------------------------------------|----------------|-----------|
| Dosilicon Co Ltd (688110.SS)                    | U (09/06/2024) | Rmb115.15 |
| Shenzhen Longsys Electronics Co Ltd (301308.SZ) | E (02/25/2026) | Rmb373.90 |

**Tiffany Yeh**

|                                                |                |              |
|------------------------------------------------|----------------|--------------|
| AllRing Tech Co. (6187.TWO)                    | O (09/23/2025) | NT\$1,035.00 |
| FOCI Fiber Optic Communications Inc (3363.TWO) | O (01/15/2025) | NT\$598.00   |
| Himax Technologies Inc (HIMX.O)                | E (02/04/2026) | US\$12.62    |
| Hon Precision (7769.TW)                        | O (04/17/2026) | NT\$6,110.00 |
| MPI Corporation (6223.TWO)                     | O (04/17/2026) | NT\$5,720.00 |
| Silicon Motion (SIMO.O)                        | O (05/06/2024) | US\$270.90   |
| WinWay Technology Co Ltd (6515.TW)             | O (04/17/2026) | NT\$6,395.00 |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.